

Formation of *O*-Allyl Oximes by Reaction of Allylzinc Reagents with α -Chloronitroso Compounds

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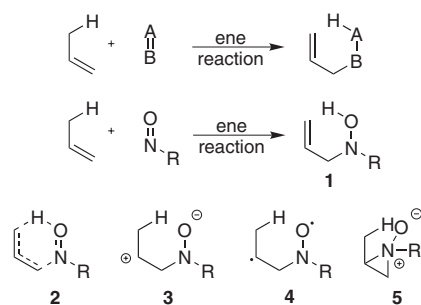
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Received 13 May 2005; revised 23 June 2005

Abstract: The reaction of α -chloronitrosocyclohexane with allylzinc reagents in THF selectively provides *O*-allylcyclohexanone oximes in very good yields in an ene type reaction followed by zinc halide elimination. *N*-Allyl hydroxylamines are formed as minor products via *N*-allyl nitrones when a nonpolar solvent, e.g. toluene, is employed. In both cases complete allylic transposition with respect to the allylic organozinc reagent is observed. Compared to non-metalated allyl compounds, which give *N*-allyl nitrones upon reaction with α -chloronitroso compounds, the use of zinc as hydrogen equivalent in an ene type reaction displays a drastic change of reactivity.

Key words: ene reactions, regioselectivity, zinc, allylations, organometallic reagents

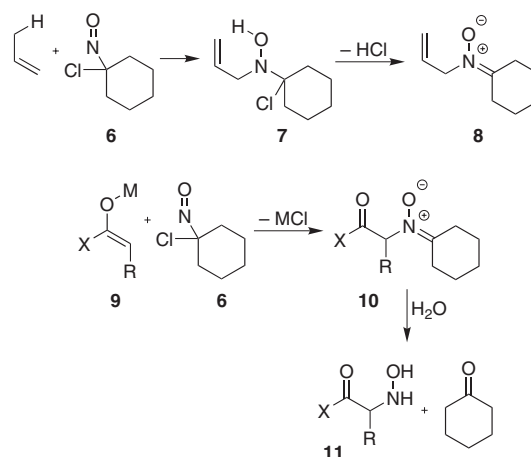
The ene reaction and its hetero variants belong to the central repertoire of chemical reactions. While the ene reaction itself routinely is considered as being concerted, the mechanisms of hetero-ene reactions are still subject of controversial discussions. Among the different hetero-enophiles $A=B$ (Scheme 1), the isoelectronic species singlet oxygen, 1,2,4-triazoline-3,5-diones, and nitroso compounds attracted special attention.² The debates on the mechanism of the nitroso ene reaction recently experienced new impact based on theoretical studies.^{2,3}



Scheme 1 Intermediates of the hetero-ene reaction

Reactions of nitroso compounds with alkenes were discovered in 1965 and give *N*-allyl hydroxylamine derivatives **1** according to an ene-type mechanism.⁴ Besides concerted transition states **2** and zwitterionic species **3**, especially polarized diradical intermediates **4**, and/or aziridine *N*-oxide intermediates **5** are currently being

discussed. While the initially formed *N*-allyl hydroxylamines **1** are susceptible to oxidation and disproportionation, such disadvantages are not observed for products resulting from reaction with α -chloronitroso compounds (e.g. **6**) as highly reactive, electron deficient enophiles.^{5,6} The primary ene products **7** eliminate HCl to give nitrones **8** (Scheme 2). Compared to α -chloronitrosoalkanes, which require days or weeks to react completely in ene reactions,^{5a} an enhanced reactivity has been observed for the even more electron-deficient 1-chloro-1-nitrosocarbonyl compounds, due to the electronegative oxygen substituent.²

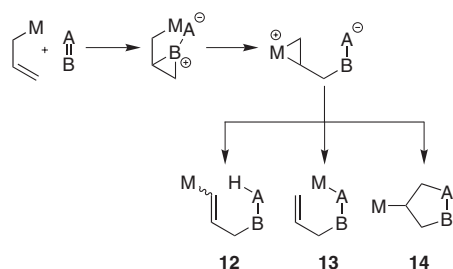


Scheme 2 Reactions of α -chloronitroso compounds **6**

N-substituted nitrones **10** are obtained as intermediates upon reaction of sodium or zinc enolates **9** ($M = ZnCl$) with α -chloronitroso compounds. Products **10** are easily hydrolyzed to give α -(*N*-hydroxyamino)ketones and α -(*N*-hydroxyamino)amides **11**, respectively.⁷ The principle of a metal substituent acting as a hydrogen equivalent in ene reactions suggests that suitable allylmetallic compounds should react with the same enophiles by transfer of the metal rather than hydrogen. The use of a metal as hydrogen equivalent in such metallo-ene reactions⁸ is important because it provides the option of modulating the reactivity by varying the metal and, if possible, its ligands.

Several examples of these metallo-ene reactions with silicon, germanium, tin, lead, or mercury as the metal and singlet oxygen or an azo compound (1,2,4-triazoline-3,5-dione) as the enophile have been established. The proposed mechanism by which metalated allyl systems react with enophiles, based on that for the non-metalated allylic

compounds, shows that the products are those which result from the H-ene reaction (**12**), the M-ene reaction (**13**), and a [2+3] cycloaddition with a 1,2-shift of the metallic group (**14**) (Scheme 3).⁹ It has also been mentioned that a reasonable model can involve prior formation of a complex between the ene and the enophile A=B.

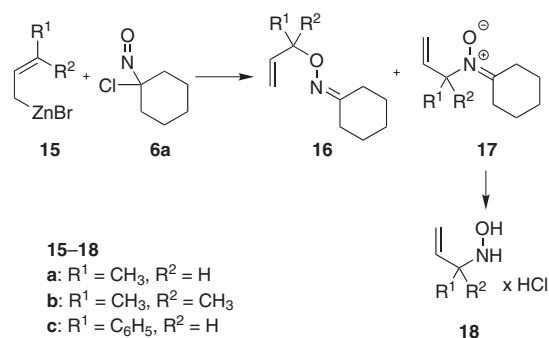


Scheme 3 Metallo-ene Reaction⁹

Until now, there are no reports concerning the reaction between allylorganometallic compounds and α -chloronitroso derivatives. Inspired by ene-type reactions with organometallic reagents and the reactivity of α -chloronitrosoalkanes with allyl systems, we investigated the reactivity of α -chloronitrosocyclohexane **6a** towards allylzinc reagents **15**.

In close analogy to the enolate reactions (Scheme 2), we expected *N*-allyl hydroxylamines **18** to be formed via intermediate nitrones **17** (Scheme 4), as in the course of a similar reaction dimethylzinc reacts with 2-chloro-2-nitrosopropane to give *N*-methylhydroxylamine.¹⁰ The reactions were performed by titrating a vigorously stirred THF solution of α -chloronitrosocyclohexane **6a** with the allylzinc reagents **15** (Scheme 4, Table 1). The end point was indicated by the disappearance of the blue color of the α -chloronitroso compound. After quenching with absolute methanol, TLC analysis of the reaction mixture indicated

formation of a single product and total conversion of **6a**. Surprisingly, *O*-allyl cyclohexanone oximes **16** were obtained exclusively upon reaction of allylzinc compounds **15** with 1-chloro-1-nitrosocyclohexane (**6a**) in THF as the solvent. No formation of the regioisomeric unbranched *O*-allyl oximes was detected.



Scheme 4 Reaction of α -chloronitrosocyclohexane **6a** with the allylzinc compounds **15**

The direct conversion of oximes to oxime ethers has been limited to the alkylation of oximes, but recently a transition metal-catalyzed allylic substitution with oximes was shown to provide *O*-allyl oxime ethers.¹¹

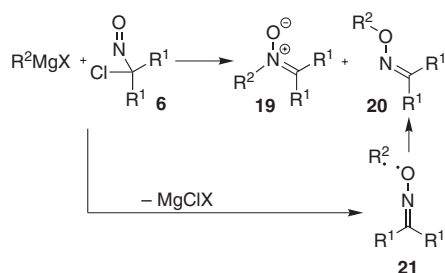
Using toluene as the solvent, oximes **16a** and **16b** were accompanied by small amounts of allyl hydroxylamine hydrochlorides **18a** and **18b** (Table 1). The latter were isolated upon quenching with methanol and acid hydrolysis of the reaction mixture. However, *O*-(1-phenylallyl)cyclohexanone oxime (**16c**) was formed exclusively from **15c** under the same reaction conditions.

de Boer et al.^{12a} reported that a complex mixture of products results from the reaction of α -chloronitrosoalkanes (e.g. 2-chloro-2-nitrosoadamantane) with Grignard re-

Table 1 Yields of *O*-Allyl Oximes **16** and Hydroxylamines **18**

Reagent	R ¹	R ²	Solvent	Temp (°C)	Yield (%) 16	Yield (%) 18
15a	Me	H	THF	−78	96 16a	–
15a	Me	H	THF	0	97 16a	–
15b	Me	Me	THF	22	95 16b	–
15b	Me	Me	THF	−78	82 16b	–
15c	Ph	H	THF	−78	84 16c	–
15a	Me	H	toluene	−78	84 16a	14 18a
15a	Me	H	toluene	0	82 16a	8 18a
15b	Me	Me	toluene	−78	72 16b	2 18b
15b	Me	Me	toluene	0	71 16b	3 18b
15c	Ph	H	toluene	−78	73 16c	–
15c	Ph	H	toluene	0	64 16c	–

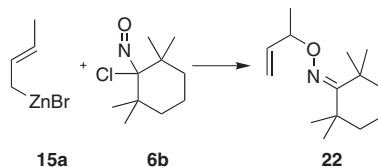
agents RMgX ($\text{R} = \text{Me}$, $n\text{-Bu}$, $t\text{-Bu}$, neopentyl, Ph , Bn , $\text{X} = \text{Br}$, I). Besides the expected nitrones **19**, which were obtained in yields between 6 and 87%, adamantanone oximes (4–60%) and adamantanone oxime ethers **20** (traces–28%) were obtained. The nitrones result from the regular 1,2-addition of the Grignard reagents across the $\text{N}=\text{O}$ group,^{12b} whereas the surprising formation of the oxime ethers **20** was accounted for by a single electron-transfer forming a relatively stable iminoxy radical **21** that would be able to recombine with the alkyl radical to give **20** (Scheme 5). Moreover, **21** also might abstract a hydrogen atom to give the oxime. Despite being mechanistically interesting, the reaction clearly does not have any preparative significance because of the different side products.



Scheme 5 Addition of Grignard reagents to α -chloronitroso compounds^{12a}

The SET from an organometallic reagent to a substrate has been suggested to be an outer sphere process not necessarily requiring intimate approach of the interacting species.^{12c} Such a process is expected to be less disfavored by steric hindrance than polar 1,2-addition of the organometallic reagent to the nitroso group. The involvement of a SET process has been proven by de Boer et al. using the highly congested **6b** as substrate, where 1,2-addition of the Grignard reagent to the nitroso group is reported to be completely hindered by the flanking methyl substituents. Steric constraints being less severe for an electron transfer, this should virtually become the exclusive process with the Grignard reagent. However, in our case, the occurrence of such a radical pathway is less probable as no free oxime or products derived from the reaction of **21** with toluene were detected. Moreover, **6b** was unreactive toward 2-butenylzinc bromide **15a** in THF or toluene at temperatures below 4 °C. Stirring at room temperature for 12 hours provided the oxime ether **22** in 31% (THF) and 28% (toluene) yield accompanied only by unreacted **15a** (Scheme 6). The formation of an unbranched *O*-allyl oxime was not observed. These observations rule out the occurrence of a radical reaction involving SET from the organozinc reagent to the α -chloronitroso compound.

Quite likely, the reaction proceeds as a metallo-hetero-ene reaction followed by zinc halide elimination from **23** yielding **16** (Schemes 4, 7a). The formation of the hydroxylamines **18** upon acidic hydrolysis of the reaction mixture, when toluene is used as solvent, sustains the formation of nitrones **17** in a metallo-hetero-ene process with

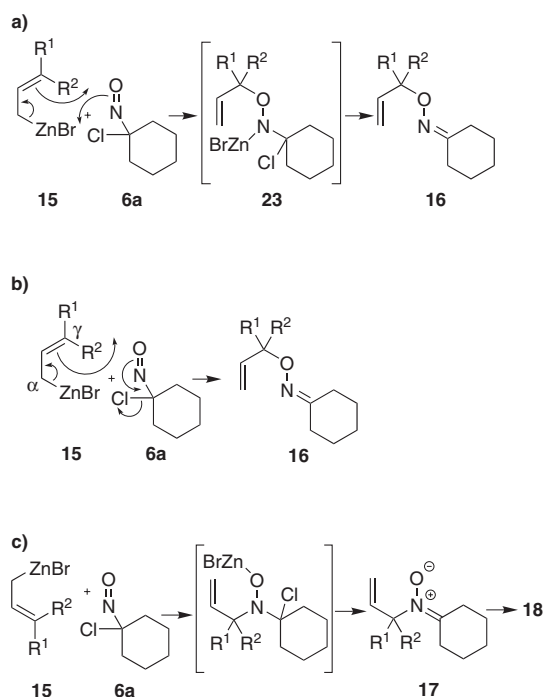


Scheme 6 Formation of oxime ether **22** from sterically congested **6b**

inverted orientation (Scheme 7c). The unpolar solvent toluene presumably enhances the Lewis acid-base interaction between zinc and oxygen, which might favor the *M*-ene process yielding **17**. Alternatively, **16** could be formed by γ -attack of the allylzinc compound **15** to the 'hetero-allyl' halide **6a** (Scheme 7b), where the halide leaving group permits attack to the nitroso oxygen and, hence, explains the orientation observed. However, other metalated compounds preferably (RLi , RMgX) or exclusively (Zn enolates) attack the nitroso nitrogen of **6** (Schemes 2, 5).^{7,12} This finding contradicts the interpretation that the special substitution pattern of α -chloronitroso compounds like **6** would be responsible for the regiochemistry and supports the hypothesis of a metallo-hetero-ene reaction with two possible enophile orientations.

The absence of nitron **17**, and consequently hydroxylamine hydrochloride **18c** (Table 1), when 3-phenylallylzinc bromide (**15c**) was reacted with **6a**, is most probably due to the greater steric hindrance caused by the bulkiness of the phenyl moiety ($\text{R}^1 = \text{Ph}$, Scheme 4).

In summary, the reaction of α -chloronitrosocyclohexane **6a** with allylzinc reagents in THF provides *O*-allyl cyclohexanone oximes in very good yield and high regioselectivity.



Scheme 7 Regiochemistry and mechanistic considerations for the reaction

tivity, also with respect to the allyl moiety. Minor amounts of *N*-allyl hydroxylamines are also formed when a nonpolar solvent, e.g. toluene, is employed. In both cases, complete allylic transposition with respect to the allylic organozinc reagent (γ -attack of the allylzinc compound) is observed. Hence, the reaction of α -chloronitroso compounds with allyl zinc reagents is synthetically more useful than with Grignard reagents, where many by-products were observed.

The reaction outcome may be rationalized by metallo-hetero-ene reactions with different orientations, followed by zinc halide elimination. Compared to non-metalated allyl compounds, which provide *N*-allyl nitrones when reacting with α -chloronitroso compounds, the use of zinc as hydrogen-equivalent in the ene reaction leads to a drastic change in reactivity.

All reactions were carried out in flame-dried glassware under dry oxygen-free N_2 . But-2-enyl, 3,3-dimethylallyl and 3-phenylallyl bromides were purchased from Fluka and used without further purification. Zinc powder (granulation <63 μ m) was purchased from Fluka. Organometallic reagents **15** were prepared as reported¹³ from the corresponding bromides and activated zinc^{14,15} in THF. The concentration of the organozinc solutions was determined by quenching an aliquot with I_2/Et_2O and GC analysis, using *n*-dodecane as internal standard, and is given in mol/L.¹⁴ Toluene solutions of **15** were prepared by evaporation of THF in vacuo, followed by addition of anhyd toluene under N_2 . Et_2O and THF were distilled immediately before use from sodium benzophenone ketyl. Toluene was successively distilled over CaH_2 and sodium prior to use. Petroleum ether refers to the fraction with bp 35–60 °C. α -Chloronitroso cyclohexane **6a** was prepared as reported¹⁶ and stored at –20 °C. 1-Chloro-1-nitroso-2,2,6,6-tetramethylcyclohexane (**6b**) was prepared from 2,2,6,6-tetramethylcyclohexanone oxime. An improved method, starting from 2,2,6,6-tetramethylcyclohexanone (see below), was used for oxime synthesis.¹⁷

Melting points were determined using a Büchi Melting Point B-540 apparatus and are uncorrected. IR spectra were recorded using a Jasco FT/IR-410 spectrometer. 1H and ^{13}C NMR data were recorded with a Bruker DRX 500 spectrometer at 500 MHz, respectively 125 MHz. Chemical shifts δ are given in ppm relative to TMS; coupling constants *J* are reported in Hz. GC-MS analysis was performed on Shimadzu GC-QP5050A (GC: 30-m HP5-MS capillary column, 0.33 μ m, helium as the carrier gas, temperature program: 75 °C, 5 min, 75 °C $\rightarrow$ 270 °C @ 15 °C/min, 270 °C, 5 min, MS: EI, ionization energy: 70 eV). MS spectra were recorded using an Autospec X magnetic sector mass spectrometer with EBE geometry (Vacuum Generators, Manchester, UK) equipped with a standard EI/CI source. Peaks are reported as *m/z* (relative intensity in %). Elemental analyses were performed with a CHNS-932 (Leco Corporation) system. Analytical TLC plates (Silica gel 60 F₂₅₄ on aluminum foil) were purchased from Merck. Silica gel 60 (40–63 μ m, Merck) was used for flash chromatography.

Reaction of Organozinc Reagents **15** with α -Chloronitroso-cyclohexane **6a** in THF; General Procedure

A solution of α -chloronitrosocyclohexane **6a** (1.00 g, 6.77 mmol) in anhyd THF (5 mL) was cooled to –78 °C and the THF solution of the organozinc reagent **15** (20 mL, 6.77 mmol) was added dropwise via syringe with stirring until the blue color had disappeared. The resulting light-yellow solution was quenched with absolute MeOH (1 mL) and left to reach r.t. TLC analysis showed formation of a single product and total conversion of **6a**. *tert*-Butyl methyl ether (10 mL) was added and after washing with sat. NH_4Cl solution (20

mL), the organic phase was dried (Na_2SO_4) and the solvents were evaporated in vacuo.

Reaction of Organozinc Reagents **15** with α -Chloronitroso-cyclohexane **6a** in Toluene; General Procedure

A solution of α -chloronitrosocyclohexane **6a** (0.5 g, 3.39 mmol) in anhyd toluene (5 mL) was cooled to –78 °C and the toluene solution of the organozinc reagent **15** (10 mL, 3.39 mmol) was added dropwise via syringe with stirring until the blue color had disappeared. The resulting light-yellow solution was quenched with absolute MeOH (1 mL) and left to reach r.t. After evaporation of solvents in vacuo, the residue was dissolved in $CHCl_3$ (20 mL) and extracted with a solution of 1 M HCl (50 mL). GC analysis of the organic phase showed the formation of cyclohexanone as the hydrolysis product of intermediary nitrones. Flash chromatography on silica gel (*t*-BuOMe–petroleum ether, 1:10) of the organic phase afforded analytically pure oximes **16**. The collected aqueous phases were neutralized with $KHCO_3$, extracted with Et_2O and dried (Na_2SO_4). The hydroxylamine hydrochlorides **18** were isolated by precipitation from Et_2O solution using gaseous HCl (Table 1).

O-(But-1-en-3-yl)cyclohexanone Oxime (**16a**)

Colorless oil; *R*_f 0.57 ($EtOAc$ –petroleum ether, 2:5).

IR (neat): 3081, 2981, 2932, 2859, 1642, 1448, 1371, 1239, 1134, 945 cm^{-1} .

1H NMR ($CDCl_3$): δ = 5.91 (ddd, 1 H, *J* = 17.3, 10.6, 5.8 Hz), 5.19 (ddd, 1 H, *J* = 17.3, 1.5, 1.4 Hz), 5.10 (ddd, 1 H, *J* = 10.6, 1.5, 1.4 Hz), 4.59 (dtq, 1 H, *J* = 5.8, 1.4, 6.4 Hz), 2.45–2.50 (m, 2 H), 2.17–2.22 (m, 2 H), 1.56–1.68 (m, 6 H), 1.29 (d, 3 H, *J* = 6.4 Hz).

^{13}C NMR ($CDCl_3$): δ = 160.0, 140.2, 114.7, 78.5, 32.3, 27.2, 25.9, 25.8, 25.4, 19.8.

MS (GC-MS) *m/z* (%) = 167 (14), 152 (10), 113 (14), 96 (17), 85 (19), 55 (100).

Anal. Calcd for $C_{10}H_{17}NO$ (167.25): C, 71.81; H, 10.25; N, 8.37. Found: C, 71.97; H, 10.16; N, 8.12.

O-(3-Methylbut-1-en-3-yl)cyclohexanone Oxime (**16b**)

Colorless oil; *R*_f 0.52 (*t*-BuOMe–petroleum ether, 1:10).

IR (neat): 3084, 2987 (CH), 2932 (CH_2), 2859 (CH_3), 1642 (C=N), 1449 (CH_2), 1373 (CH_3), 1253 (N–O), 1153 (C–O), 945 cm^{-1} (C=CH).

1H NMR (CD_2Cl_2): δ = 6.01 (dd, 1 H, *J* = 17.5, 10.6 Hz), 5.09 (dd, 1 H, *J* = 17.5, 1.5 Hz), 5.00 (dd, 1 H, *J* = 10.9, 1.5 Hz), 2.42–2.45 (m, 2 H), 2.13–2.16 (m, 2 H), 1.55–1.63 (m, 6 H), 1.31 (s, 6 H).

^{13}C NMR (CD_2Cl_2): δ = 159.2, 145.4, 112.2, 79.0, 32.7, 27.6, 26.3, 26.2, 25.9, 25.5.

MS (GC-MS): *m/z* (%) = 181 (5), 166 (2), 151 (3), 114 (7), 69 (100), 55 (5), 41 (55).

Anal. Calcd for $C_{11}H_{19}NO$ (181.27): C, 72.88; H, 10.56; N, 7.73. Found: C, 72.55; H, 10.63; N, 7.45.

O-(1-Phenylallyl)cyclohexanone Oxime (**16c**)

Light yellow oil; *R*_f 0.43 (*t*-BuOMe–petroleum ether, 1:5).

IR (neat): 3084, 3062, 3029, 2982, 2932, 2858, 1641, 1449, 1346, 1254, 1026, 990, 931, 918, 888, 700 cm^{-1} .

1H NMR ($CDCl_3$): δ = 7.22–7.33 (m, 5 H), 6.06 (ddd, 1 H, *J* = 17.2, 10.6, 6.2 Hz), 5.54 (dt, 1 H, *J* = 6.2, 1.2 Hz), 5.24 (dd, 1 H, *J* = 17.2, 1.2 Hz), 5.20 (dd, 1 H, *J* = 10.6, 1.2 Hz), 2.52–2.57 (m, 2 H), 2.15–2.19 (m, 2 H), 1.51–1.70 (m, 6 H).

^{13}C NMR ($CDCl_3$): δ = 161.0, 140.8, 138.4, 128.2, 127.4, 127.0, 116.2, 84.8, 32.1, 27.0, 25.8, 25.7, 25.6.

MS (GC-MS): m/z (%) = 230 (2), 117 (100), 115 (20), 105 (5), 91 (10), 77 (5).

Anal. Calcd for $C_{15}H_{19}NO$ (229.32): C, 78.56; H, 8.35; N, 6.11. Found: C, 78.40; H, 8.33; N, 6.12.

N-(But-1-en-3-yl)hydroxylamine Hydrochloride (18a)

Mp 58–61 °C.

IR (KBr): 3465, 3040, 2507, 1633, 1450, 1426, 1381, 1003, 947 cm^{-1} .

1H NMR (DMSO- d_6): δ = 10.39 (br, 2 H), 5.84 (ddd, 1 H, J = 17.4, 10.5, 7.4 Hz), 5.32 (ddd, 1 H, J = 17.4, 2.4, 1.2 Hz), 5.28 (ddd, 1 H, J = 10.5, 2.4, 1.2 Hz), 3.68 (dtq, 1 H, J = 7.4, 1.2, 6.6 Hz), 1.14 (d, 3 H, J = 6.6 Hz).

^{13}C NMR (DMSO- d_6): δ = 136.4, 123.2, 60.0, 14.8.

MS: m/z (%) = 87 (7), 72 (35), 55 (60), 41 (100), 40 (69), 39 (42).

N-(3-Methylbut-1-en-3-yl)hydroxylamine Hydrochloride (18b)

Mp 60–63 °C.

IR (KBr): 3467, 3037, 2509, 1635, 1448, 1426, 1382, 1010, 948 cm^{-1} .

1H NMR (DMSO- d_6): δ = 10.41 (br, 2 H), 6.03 (dd, 1 H, J = 17.5, 10.6 Hz), 5.35 (d, 1 H, J = 17.5 Hz), 5.31 (d, 1 H, J = 10.6 Hz), 1.36 (s, 6 H).

^{13}C NMR (DMSO- d_6): δ = 137.2, 117.5, 61.6, 21.1.

MS: m/z (%) = 101 (6), 86 (11), 84 (5), 74 (8), 69 (53), 46 (15), 41 (100), 40 (30), 39 (25).

2,2,6,6-Tetramethylcyclohexanone Oxime

A solution of $NH_2OH \cdot HCl$ (6.77 g, 97.4 mmol) and NaOAc (8.52 g, 103.87 mmol) in MeOH (120 mL) was stirred for 30 min at r.t. 2,2,6,6-Tetramethylcyclohexanone (5.00 g, 32.46 mmol) was added and the mixture was refluxed for 3 days. After completion, the mixture was poured into ice water, the white precipitate was collected by filtration and washed with cold water. Recrystallization from MeOH afforded analytically pure 2,2,6,6-tetramethylcyclohexanone oxime as white crystals; yield: 4.00 g (73%); mp 154–155 °C.

IR (KBr): 3304, 2930, 2866, 1646, 1561, 1459, 1381, 1360, 1220, 935 cm^{-1} .

1H NMR ($CDCl_3$): δ = 8.74 (br, 1 H), 1.60–1.65 (m, 2 H), 1.51–1.56 (m, 4 H), 1.36 (s, 6 H), 1.21 (s, 6 H).

^{13}C NMR ($CDCl_3$): δ = 168.6, 40.5, 37.9, 37.6, 37.0, 30.5, 26.7, 17.4.

MS: m/z (%) = 169 (10), 154 (12), 152 (32), 141 (7), 137 (10), 126 (8), 100 (35), 87 (19), 69 (100), 55 (60), 41 (63), 27 (20).

Anal. Calcd for $C_{10}H_{19}NO$ (169.26): C, 70.96; H, 11.31; N, 8.28. Found: C, 70.93; H, 11.12; N, 8.21.

1-Chloro-1-nitroso-2,2,6,6-tetramethylcyclohexane (6b)

A solution of *t*-BuOCl (3.1 g, 20 mmol, 70% w/w in *t*-BuOH) in anhyd CH_2Cl_2 (30 mL) was added dropwise during 30 min under N_2 and protection against light to a pre-cooled (0 °C) solution of 2,2,6,6-tetramethylcyclohexanone oxime (3.05 g, 18 mmol) in anhyd CH_2Cl_2 (50 mL). After stirring for 1.5 h at 0 °C, the mixture was warmed-up to r.t. and the solvent was evaporated in vacuo. The blue residue was recrystallized from MeOH to afford **6b** as blue crystals; yield: 3.23 g (88%); mp 115–117 °C.

IR (KBr): 2977, 2939, 2872, 1578, 1471, 1386, 1370, 1202, 627 cm^{-1} .

1H NMR (CD_2Cl_2): δ = 2.71 (m, 2 H), 2.38 (m, 1 H), 2.11 (m, 1 H), 1.94 (m, 2 H), 1.34 (s, 6 H), 0.21 (s, 6 H).

^{13}C NMR (CD_2Cl_2): δ = 136.3, 42.1, 40.2, 27.5, 25.7, 19.9.

MS: m/z (%) = 168 (6), 160 (4), 158 (13), 145 (14), 143 (43), 137 (53), 133 (4), 131 (13), 123 (25), 107 (18), 105 (19), 103 (53), 97 (15), 95 (37), 91 (15), 83 (24), 81 (38), 79 (18), 77 (25), 69 (100), 67 (37), 57 (35), 55 (45), 43 (25), 41 (72).

Anal. Calcd for $C_{12}H_{18}ClNO$ (203.71): C, 58.96; H, 8.61; N, 6.88. Found: C, 58.94; H, 8.66; N, 6.81.

Reaction of But-2-enylzinc Bromide (15a) with 1-Chloro-1-nitroso-2,2,6,6-tetramethylcyclohexane (6b)

In THF: A solution of 1-chloro-1-nitroso-2,2,6,6-tetramethylcyclohexane (**6b**; 0.35 g, 1.71 mmol) in anhyd THF (10 mL) was cooled to –78 °C and a THF solution of but-2-enylzinc bromide (**15a**; 5 mL, 1.71 mmol) was added dropwise via syringe. The mixture was stirred at –78 °C for 1 h, allowed to reach 4 °C during 2.5 h with TLC and GC monitoring, and then stirred at r.t. for 12 h. The slightly blue solution was quenched with aq sat. $NaHCO_3$ solution (5 mL) and the white precipitate formed was dissolved by adding aq sat. NH_4Cl solution (10 mL). Separation of the organic phase, drying (Na_2SO_4) and evaporation of solvent in vacuo afforded *O*-(but-1-en-3-yl)-2,2,6,6-tetramethylcyclohexanone oxime **22** (0.12 g, 31%) as a colorless oil. A sample of **6b** in THF was run in parallel to monitor its potential photochemical decomposition.¹⁸

In toluene: A similar procedure as above was followed for the reaction of but-2-enylzinc bromide (**15a**) with 1-chloro-1-nitroso-2,2,6,6-tetramethylcyclohexane (**6b**), with the exception that a toluene solution of the organozinc reagent was used; yield: 28%.

O-(But-1-en-3-yl)-2,2,6,6-tetramethylcyclohexanone Oxime (22)

Colorless oil.

IR (neat): 3082, 2959, 2930, 2867, 1645, 1464, 1382, 1362, 1232, 1160, 946 cm^{-1} .

1H NMR (CD_2Cl_2): δ = 5.92 (ddd, 1 H, J = 17.3, 10.6, 5.7 Hz), 5.18 (ddd, 1 H, J = 17.3, 1.5, 1.4 Hz), 5.07 (ddd, 1 H, J = 10.6, 1.5, 1.4 Hz), 4.51 (dtq, 1 H, J = 5.7, 1.4, 6.7 Hz), 1.56–1.60 (m, 2 H), 1.47–1.50 (m, 4 H), 1.28 (s, 3 H), 1.27 (d, 3 H, J = 6.7 Hz), 1.26 (s, 3 H), 1.16 (s, 3 H), 1.14 (s, 3 H).

^{13}C NMR (CD_2Cl_2): δ = 167.1, 140.7, 114.4, 79.5, 41.4, 38.5, 38.3, 37.1, 31.1, 31.1, 27.5, 27.5, 19.9, 18.0.

MS: m/z (%) = 223 (5), 208 (4), 169 (18), 152 (25), 100 (18), 87 (8), 69 (75), 55 (100), 41 (30), 29 (12), 27 (20).

Anal. Calcd for $C_{14}H_{25}NO$ (223.35): C, 75.28; H, 11.28; N, 6.27. Found: C, 75.22; H, 11.14; N, 6.20.

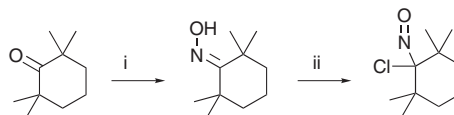
Acknowledgment

This project was supported by Deutsche Forschungsgemeinschaft and Fonds der Chemischen Industrie. S.V.F. was awarded a Ph.D. grant from Graduiertenkolleg 378/I *Mechanistische und Anwendungsaspekte nichtkonventioneller Oxidationsreaktionen*, University of Leipzig, which is gratefully acknowledged.

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Scheme 8

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